



SALAS O'BRIEN

3200 WINDY HILL ROAD SE
SUITE 200E
ATLANTA, GEORGIA 30339
PHONE 404-881-5300

Project: Fern Parc; Richardson Housing Group
Location: Norcross, GA
Description: Evergreen TH - A, Slab, Left End Unit, Right-Garage

Date: October 1, 2025
By: JTH / TJH
House Faces: Worst East

These Loads are dependent on the information/assumptions listed below. **Any deviation from these assumptions during construction could have a significant impact on the cooling and heating loads for this house and render the recommended equipment sizing invalid.** If any of these assumptions are incorrect, please let us know immediately so we can re-calculate them. This information will be used in a residential heating and cooling load program based on calculations recommended by ASHRAE, Handbook of Fundamentals. Equipment sizing indicated in this report is valid only upon verification that actual construction is in full compliance with the descriptions below and independent testing of HVAC systems has been performed; such verification and testing shall be performed by Vallus.

BUILDING DATA	
Typical Ceiling Height	9.0 ft
Floor Area	1,972 ft ²
Volume	18,518 ft ³
Percentage Glass	10 %
Number of Bedrooms	4

DESIGN CONDITIONS		
	Summer	Winter
Outdoor	95 °Fdb	18 °Fdb
Indoor	75 °Fdb	72 °Fdb
ΔT	20 °Fdb	54 °Fdb
Range	17.3 °Fdb	

ENERGY CALCULATIONS	
Right Choice HVAC kWh	7197
SEER	15
ACH50	5.0
Mechanical Ventilation	NO
House Outside Air CFM	N/A

CONSTRUCTION DESCRIPTIONS						
Walls / Partitions	#	U-Value	Description/Notes			
	W1	0.077	Above Grade	2x4	R-13 batt	
	W2	0.067	Basement	Concrete + 2x4	R-13 batt	
	W3	0.053	Attic	2x6	R-19 batt	
Doors	#	U-Value	Description/Notes			
	D1	0.2	Exterior Door		R-5	
	D2	0.5	Garage Door		R-2	
Roofs	#	U-Value	Description/Notes			
	R1	0.026	Flat Ceiling	2x4	R-38 blown	
	R2	0.026	Vaulted	2x4	R-38 blown	
Floors	#	U-Value	Description/Notes			
	F1	0.053	Framed		R-19 batt	
	F2	0.1	Slab		None	
Glass	#	Glass Type	Windows	U-Value	Internal Shading	Notes
	G1	Double Insulated	All types	0.32	Blinds/Draperies	SHGC 0.22
	G2					
	G3					

LOAD SUMMARY					
		Unit 1 - Whole House			
		Cooling		Heating	
Sensible Load		24156 Btu/hr		24250 Btu/hr	
Minimum Nominal Capacity		2.68 Tons		7.11 kW	
Load Density		735 SF/Ton		3.6 W/SF	
Floor Area Served		1972.0 SF		1972.0 SF	
Mechanical Ventilation		N/A CFM		N/A CFM	
System Size		3.0 Tons		10.0 kW	
Leaving Air Temperature		57.2 deg F		96.3 deg F	
Installed Capacity		657 SF/Ton		5.1 W/SF	

I understand the information provided herein, and warrant, to the best of my ability and knowledge, and to the full extent of my control as the Contractor, that this house will be built in accordance with the information given above.

Signed _____ Title _____ Date _____

Date: October 1, 2025
By: JTH / TJH
House Faces: Worst East

[illegible][illegible]

Unit 1 (Continued)				
Second Floor Airflows				
Room	CFM Cool	CFM Heat	CFM Avg	Duct Size
Stairs	20	35	28	
Hall	32	33	32	
Laundry	44	8	26	
Bath 2	14	17	15	
Bed 3	127	119	123	
Bed 3 WIC	44	47	45	
Bed 2 WIC	10	12	11	
Bed 2	97	111	104	
Bed 4	66	68	67	
Bed 4 WIC	5	4	5	
Primay WIC	25	46	35	
Primary Bath	37	63	50	
Primary Bed	141	101	121	
-	-	-	-	

Date: October 1, 2025
By: JTH / TJH
Level(s): First Floor
SEER 15

Cooling Size	1.5	DX Cooling kWh	1347
Heating Size	1.5	RC Heating kWh	1870
Best Case	1.5	Total kWh	3217

COOLING LOAD CALCULATION							Ceiling Height		9		9		9		9		9		9		9		9		9		9		9			
Components					CLTD (°F)	U-Value (Btu/hr-sq-F)	Cooling Load Factor	Entire House		Foyer		Family		Dining		Kitchen		Pantry		Powder												
	Type	Description	Qty	Btu/hr				Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty
Glass	G1	North	Right	(E)		7	Btu/hr-sf	0	0		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
	G1	NorthEast		(SE)		21	Btu/hr-sf	0	0		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	G1	East	Front	(S)		32	Btu/hr-sf	0	0		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	G1	SouthEast		(SW)		26	Btu/hr-sf	0	0		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	G1	South	Left	(W)		19	Btu/hr-sf	62	1153		-	54	1,004	8	149		-	-	-	-	-	-	-	-	-	-	-	-	-			
	G1	SouthWest		(NW)		29	Btu/hr-sf	0	0		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	G1	West	Back	(N)		37	Btu/hr-sf	37	1366		-	-	-	28.36	1,037	9	329		-	-	-	-	-	-	-	-	-	-	-			
	G1	NorthWest		(NE)		26	Btu/hr-sf	0	0		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	G1	Sky Light				52	Btu/hr-sf	0	0		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
G1	Shaded				7	Btu/hr-sf	12	85	12	85		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
Door	D1	Door		21	0.20	4.20	Btu/hr-sf	16	69	12	50		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
	D2	Door		30	0.50	15.00	Btu/hr-sf	18	267	17.81	267		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
Above Grade Walls	W1	North		13	0.08	1.00	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
	W1	NorthEast		19	0.08	1.46	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	East		23	0.08	1.77	Btu/hr-sf	25	45	25	45	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	SouthEast		21	0.08	1.62	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	South		16	0.08	1.23	Btu/hr-sf	390	480	145	179	110	136	135	166	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	SouthWest		21	0.08	1.62	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	West		23	0.08	1.77	Btu/hr-sf	195	345	-	-	-	-	70	125	124	220	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	NorthWest		19	0.08	1.46	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	North		13	0.08	1.00	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
Basement Walls	W1	NorthEast		19	0.08	1.46	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
	W1	East		23	0.08	1.77	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	SouthEast		21	0.08	1.62	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	South		16	0.08	1.23	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	SouthWest		21	0.08	1.62	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	West		23	0.08	1.77	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	NorthWest		19	0.08	1.46	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	North		13	0.08	1.00	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	NorthEast		19	0.08	1.46	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
Other	W1	East		23	0.08	1.77	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
	W1	SouthEast		21	0.08	1.62	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	South		16	0.08	1.23	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	SouthWest		21	0.08	1.62	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
Roof	W1	West		23	0.08	1.77	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	NorthWest		19	0.08	1.46	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W2	Grade		0	0.07	0.00	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	Partition		10	0.08	0.77	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
Floor	W1	Garage		30	0.08	2.31	Btu/hr-sf	313	723	125	290	150	347	-	-	-	-	38	87	-	-	-	-	-	-	-	-	-	-	-		
	W3	Attic		55	0.05	2.92	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W3	Kneewall		55	0.05	2.92	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	R1	Ceiling		55	0.03	1.43	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	R2	Vaulted		55	0.03	1.43	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	F1	Crawl/Uncon.		11	0.05	0.58	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	F1	Garage		30	0.05	1.59	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	F2	Slab		0	0.10	0.00	Btu/hr-sf	745	0	71	-	294	-	150	-	181	-	20	-	29	-	-	-	-	-	-	-	-	-	-		
Vent	People					255	Btu/hr	1	255			1	255																			
	Lights (W)					3.413	Btu/hr-W	374	1276	36	123	147	502	75	256	91	311	10	34	15	51	-	-	-	-	-	-	-	-			
	Appliances (W)					3.413	Btu/hr-W	930	3174			250	853			680	2,321					-	-	-	-	-	-	-	-			
	Infiltration		0.25	ACH		0.09	Btu/hr-cf				58		238		122		147		16		23		-	-	-	-	-	-	-	-		
Area	Volume				CF		6705		639		2646		1350		1629		180		261		0		0		0		0		0			
	Area				SF		745		71		294		150		181		20		29													
Subtotal (Btu/hr)								9841		1096		3334		1872		3327		50		161		0		0		0		0		0		
Safety Factor								984																								
Sensible Cooling Load (Btu/hr)								10825																								
Nominal Tons Required (75% sens. cap.)								1.20																								
System Selection 1.5 Tons								600		67		203		114		203		3		10		-		-		-		-		-		
Space CFM																																
HEATING LOAD CALCULATION																																
Components							Heating Load Factor		Entire House		Foyer		Family		Dining		Kitchen		Pantry		Powder		0		0		0		0		0	
	Type	Description	U	A	V	SF		Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	
Glass	G1		54	0.32		17.3	Btu/hr-sf	111	1924	12	207	54	933	36.36	628	9	156	0	-	0	-	0	-	0	-	0	-	0	-	0	-	
	G2		54	0.00		0.0	Btu/hr-sf	0	0	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	
	G3		54	0.00		0.0	Btu/hr-sf	0	0	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	
Door	D1		54	0.20		10.8	Btu/hr-sf	16	178	12	130	0	-	4.452	48	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	
	D2		54	0.50		27.0	Btu/hr-sf	18	481	17.81	481	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	
Walls	W1		54	0.08		4.2	Btu/hr-sf	610	2535	170.2	708	110.2	458	205	852	124.3	517	0	-	0	-	0	-	0	-	0	-	0	-	0	-	
	W2		54	0.07		3.6	Btu/hr-sf	0	0	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	
	W3		54	0.05		2.9	Btu/hr-sf	0	0	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	
	W1	Partition		27	0.08		2.1	Btu/hr-sf	313	651	125.5	261	150	312	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-
Floor</																																

Date: October 1, 2025
By: JTH / TJH
Level(s): Second Floor
SEER 15

Cooling Size	1.5	DX Cooling kWh	1659
Heating Size	1.5	RC Heating kWh	2321
Best Case	1.5	Total kWh	3980

COOLING LOAD CALCULATION							Ceiling Height		19		9		9		9		9		9		9		9		9		9		9		9		9		9			
Components					CLTD (°F)	U-Value (Btu/hr-sq-F)	Cooling Load Factor	Entire House		Stairs		Hall		Laundry		Bath 2		Bed 3		Bed 3 WIC		Bed 2 WIC		Bed 2		Bed 4		Bed 4 WIC		Primay WIC		Primary Bath		Primary Bed				
	Type	Description						Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty
Glass	G1	North	Right	(E)		7 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
	G1	NorthEast		(SE)		21 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	G1	East	Front	(S)		32 Btu/hr-sf	60	1922	-	-	-	-	-	30	961	15	480	-	-	15	480	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
	G1	SouthEast		(SW)		26 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	G1	South	Left	(W)		19 Btu/hr-sf	15	279	-	-	-	-	-	-	-	-	-	-	-	-	-	-	15	279	-	-	-	-	-	-	-	-	-	-	-	-		
	G1	SouthWest		(NW)		29 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	G1	West	Back	(N)		37 Btu/hr-sf	30	1097	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	30	1,097	-	-	-			
	G1	NorthWest		(NE)		26 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	G1	Sky Light				52 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
G1	Shaded				7 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
Door	D1	Door		21	0.20	4.20 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
	D2	Door		55	0.50	27.50 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
Above Grade Walls	W1	North		13	0.08	1.00 Btu/hr-sf	30	30	-	-	-	-	30	30	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
	W1	NorthEast		19	0.08	1.46 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	East		23	0.08	1.77 Btu/hr-sf	178	315	-	-	-	-	92	162	54	96	-	-	32	56	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	SouthEast		21	0.08	1.62 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	South		16	0.08	1.23 Btu/hr-sf	507	624	-	-	-	-	-	-	30	37	-	-	178	219	106	130	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	SouthWest		21	0.08	1.62 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	West		23	0.08	1.77 Btu/hr-sf	207	366	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-				
	W1	NorthWest		19	0.08	1.46 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	North		13	0.08	1.00 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
Basement Walls	W1	NorthEast		19	0.08	1.46 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	East		23	0.08	1.77 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	SouthEast		21	0.08	1.62 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	South		16	0.08	1.23 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	SouthWest		21	0.08	1.62 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	West		23	0.08	1.77 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	NorthWest		19	0.08	1.46 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W2	Grade		0	0.07	0.00 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	Partition		10	0.08	0.77 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
Other	W1	Garage		30	0.08	2.31 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
	W3	Attic		55	0.05	2.92 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W3	Kneseall		55	0.05	2.92 Btu/hr-sf	26	75	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	R1	Ceiling		55	0.03	1.43 Btu/hr-sf	1023	1463	77	110	131	187	24	34	46	66	165	236	33	47	32	46	161	230	139	199	22	31	76	109	104	149	13	19	-	-		
	R2	Vaulted		55	0.03	1.43 Btu/hr-sf	288	413	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	F1	Cantilver		20	0.05	1.06 Btu/hr-sf	14	15	-	-	-	-	-	-	-	-	-	-	-	-	-	14	15	-	-	-	-	-	-	-	-	-	-	-	-			
	F1	Garage		30	0.05	1.59 Btu/hr-sf	409	650	-	-	-	-	39	62	21	33	46	73	165	262	33	52	32	51	73	116	-	-	-	-	-	-	-	-	-			
	F2	Slab		0	0.10	0.00 Btu/hr-sf	58	0	58	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	Internal	People					255 Btu/hr	4	1020	-	-	-	-	-	-	1	255	-	-	1	255	1	255	1	255	-	-	-	-	-	-	-	-	1	255	-	-	
Lights (W)						3.413 Btu/hr-W	617	2106	39	133	66	225	12	41	23	78	83	283	17	58	16	55	81	276	70	239	11	38	38	130	52	177	109	372	-	-		
Appliances (W)						3.413 Btu/hr-W	200	683	-	-	-	200	683	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
Infiltration			0.25	ACH		0.09 Btu/hr-cf		1063		132		106		19		37		134		27		26		130		113		18		62		84		176	-	-		
Volume					CF	11813		1463		1179		216		414		1485		297		1485		288		1449		1251		198		684		936		1953		0		
Area					SF	1227		77		131		24		46		165		33		32		161		139		122		76		104		217		0	-			
Subtotal (Btu/hr)							12119		375		581		811		255		2324		798		177		1778		1215		87		457		685		2578		0			
Safety Factor							10 %																															
Sensible Cooling Load (Btu/hr)							13331																															
Nominal Tons Required (75% sens. cap.)							1.48																															
System Selection 1.5 Tons Space CFM							600		19		29		40		13		115		39		9		88		60		4		23		34		128		-			
HEATING LOAD CALCULATION							Ceiling Height		19		9		9		9		9		9		9		9		9		9		9		9		9		9			
Components					CLTD (°F)	U-Value (Btu/hr-sq-F)	Heating Load Factor	Entire House		Stairs		Hall		Laundry		Bath 2		Bed 3		Bed 3 WIC		Bed 2 WIC		Bed 2		Bed 4		Bed 4 WIC		Primay WIC		Primary Bath		Primary Bed		0		
	Type	Description						Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty
Glass	G1			54	0.32	17.3 Btu/hr-sf	105	1814	0	-	0	-	0	-	0	-	30	518	15	259	0	-	15	259	15	259	0	-	0	-	0	-	0	-	30	518	0	-
	G2			54	0.00	0.0 Btu/hr-sf	0	0	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-		
	G3			54	0.00	0.0 Btu/hr-sf	0	0	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-		
Door	D1			54	0.20	10.8 Btu/hr-sf	0	0	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-		
	D2			54	0.50	27.0 Btu/hr-sf	0	0	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-		
Walls	W1			54	0.08	4.2 Btu/hr-sf	921	3829	0	-	0	-	0	-	0	-	121.																					